

P0306

Diagnostic information - DME (DFI) control unit

Misfiring

INFORMATION 1

The entry **P13A3** (Misfiring that is damaging to catalytic converter with cylinder deactivation) in the fault memory is for information purposes only.

♦ It is stored in order to track the behavior of the vehicle.

INFORMATION 2

The entry **P0313** (Misfiring when fuel tank level is low) in the fault memory provides additional information about the faults (P0300 ... P0306), which are also entered and indicate the cylinders on which misfiring has occurred.

► In this case, check whether misfiring may have been caused only as a result of the low fuel tank level .

Diagnostic conditions

- Engine temperature > -20 °C (-4 °F)
- Idle speed up to the red area on the tachometer
- No overrun phase(s)
- None of the following faults stored: P0335, P0336, P0371, P0373

Possible fault causes

J Affected by block heater

J Loose contact or corrosion on connectors or lines

J Power supply or ground supply faulty

J Spark plug(s) oily due to oil entrainment in the cylinder

J Wrong spark plug(s)

J Mechanical or electrical fault in spark plug(s)

J Bad ignition module faulty

J Power supply or ground supply faulty

J Wrong fuel quality

J Fuel tank empty (P0313)

J Water in fuel tank

J Fuel line dirty/blocked

J Mechanical fault in electric fuel pump

J Mechanical fault in fuel high-pressure pump

J Insufficient fuel due to foreign bodies (chips) ahead of the fuel high-pressure rail (see K

INFORMATION 3)

J Fuel injector faulty or borderline

(opens too late/not long enough/not enough or leaking)

J Oxygen sensor(s) faulty

J Exhaust flap faulty

J Exhaust system constricted/blocked
J Catalytic converter faulty

J Leak in intake-air distributor
J Foreign bodies in intake-air distributor
J Camshaft adjustment fault
J Timing wrong
J Valve lift control fault
J Valve tappets faulty (a valve has not switched to the other lift or has not switched quickly enough)
J Mechanical fault in valve drive
J Insufficient compression
(e.g. cylinder liner damaged, valve seat ring faulty, ...)
I ONLY for manual transmission:
J Clutch and flywheel damaged as a result of thermal overload

Fault setting condition(s)

- The diagnostic function detects uneven running for the specified cylinder.
 - Uneven running is calculated from the expected and actual crankshaft acceleration due to combustion.
 - Uneven running is interpreted as a misfire or valve lift fault, depending on type.
-  It is possible that only one cylinder (or the cylinders of one bank) has/have exceeded the uneven running threshold for a fault entry. Other cylinders may be just below the level which would result in a fault entry.
⇒ See actual values [M005](#) to [M060](#)

INFORMATION 3

Example

- ♦ Vehicle speed approx. 140 km/h (86 mph)
- ♦ Little or no acceleration although accelerator pedal is floored
- ♦ Actual values, or values in extended fault memory when fault was entered or when lack of power was detected, showed:
 - ♦ Fault entry P2414
 - ♦ Oxygen sensing, bank 1 = 25.0% (enrichment)
 - ♦ Fuel trim adaptation (load-dependent), bank 1 = 14.844%
 - ♦ L070_Oxygen sensor (LSU) voltage ahead of cat. con. 1 = 2.98828 V
 - ♦ L090_Oxygen sensor voltage behind cat. con. 1 = 0.00000 V

Explanation

The inlet for the fuel high-pressure rail behind the high-pressure pump is blocked either fully or partially by foreign bodies.

OR

The connection between the two fuel high-pressure rails is blocked either fully or partially by foreign bodies.

 The inlet of the respective rail is constricted by a restrictor, which is used to dampen pressure peaks.

- ♦ The fuel high-pressure sensor, which is fitted at the end of the lines, measures a pressure

increase that is too slow, or a rapid drop in pressure when the fuel injectors open.

- ♦ As a result, fuel high pressure that is too high is set in the rail AHEAD OF the blockage, while at the same time, there is insufficient pressure and volume BEHIND the blockage.
- ♦ The oxygen sensors of the affected bank measure a large oxygen surplus, while at the same time, fuel trim adaptation tries to enrich the mixture as much as possible (extension of the injection time).

Consequences

- ♦ The following faults can be entered individually, in combination, or might not be entered at all:
 - ♦ Mixture control fault
 - ♦ Load detection fault
 - ♦ Misfiring
 - ♦ Oxygen sensor fault

AND

- ♦ Lack of power in upper load range can be reproduced, but is not necessarily always present.

OR

- ♦ The engine can stall again while starting.

Cause

After removing the lines between the fuel high-pressure pump and high-pressure rail, you can sometimes see chips or hear them rattling.

P0313

Diagnostic information - DME (DFI) control unit

Misfiring

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J Bar ignition module faulty

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****i** INFORMATION 3**

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Cause

After removing the lines between the fuel high-pressure pump and high-pressure rail, you can sometimes see chips or hear them rattling.

Diagnostic information - DME (DFI) control unit

Knock sensor 1

Diagnostic conditions

- Perform knock control short test

or

- Engine running, with:
 - Time since engine start > 10 second(s)
 - Engine temperature > 50 °C (122 °F)
 - Engine speed > 1,000 rpm
 - Different load ranges

Possible fault causes

- ♦ Loose contact or corrosion on connectors or lines
- ♦ Short circuit to ground or open circuit in the lines
- ♦ Knock sensor 1 loose
- ♦ Knock sensor 1 fitted too tightly
- ♦ Knock sensor 1 faulty
- ♦ Engine oil filling level much too high

Fault setting condition(s)

- The fault is detected if the knock sensor signal does not change sufficiently or is too weak, depending on the engine speed and load, and possibly also if it is too quiet (noise is less dynamic or quieter than expected).

Fault effects

- ♦ If the fault is stored, the ignition timing is retarded for all cylinders affecting this knock sensor in the >knock control active< range.
- ♦ Knock control adaptation is idle.

Diagnostic information - DME (DFI) control unit

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- The fault is detected if the knock sensor signal does not change sufficiently or is too weak, depending on the engine speed and load, and possibly also if it is too quiet (noise is less dynamic or quieter than expected).

Fault effects

- ♦ If the fault is stored, the ignition timing is retarded for all cylinders affecting this knock sensor in the >knock control active< range.
- ♦ Knock control adaptation is idle.

Diagnostic information - DME (DFI) control unit

Knock sensor 2

Diagnostic conditions

- Perform knock control short test

or

- Engine running, with:
 - Time since engine start > 10 second(s)
 - Engine temperature > 50 °C (122 °F)
 - Engine speed > 1,000 rpm
 - Different load ranges

Possible fault causes

- ♦ Loose contact or corrosion on connectors or lines
- ♦ Short circuit to ground or open circuit in the lines
- ♦ Knock sensor 2 loose
- ♦ Knock sensor 2 fitted too tightly
- ♦ Knock sensor 2 faulty
- ♦ Engine oil filling level much too high

Fault setting condition(s)

- The fault is detected if the knock sensor signal does not change sufficiently or is too weak, depending on the engine speed and load, and possibly also if it is too quiet (noise is less dynamic or quieter than expected).

Fault effects

- ♦ If the fault is stored, the ignition timing is retarded for all cylinders affecting this knock sensor in the >knock control active< range.
- ♦ Knock control adaptation is idle.

Diagnostic information - DME (DFI) control unit

Knock sensor 2

Diagnostic conditions

- Perform knock control short test

or

- Engine running, with:
 - Time since engine start > 10 second(s)
 - Engine temperature > 50 °C (122 °F)
 - Engine speed > 1,000 rpm
 - Different load ranges

Possible fault causes

- ♦ Loose contact or corrosion on connectors or lines
- ♦ Short circuit to ground or open circuit in the lines
- ♦ Knock sensor 2 loose
- ♦ Knock sensor 2 fitted too tightly
- ♦ Knock sensor 2 faulty
- ♦ Engine oil filling level much too high

Fault setting condition(s)

- The fault is detected if the knock sensor signal does not change sufficiently or is too weak, depending on the engine speed and load, and possibly also if it is too quiet (noise is less dynamic or quieter than expected).

Fault effects

- ♦ If the fault is stored, the ignition timing is retarded for all cylinders affecting this knock sensor in the >knock control active< range.
- ♦ Knock control adaptation is idle.

P0335

Diagnostic information - DME (DFI) control unit

Crankshaft sensor

Diagnostic conditions

- Engine running
- Then automatic engine stop and restart by the Start Stop system

Possible fault causes

- ♦ Loose contact or corrosion on connectors or lines
- ♦ Short circuit to B+/ground/between lines or open circuit in the lines
- ♦ Phase sensor wheel (flywheel) faulty (damaged/bent)
- ♦ Crankshaft sensor not installed or installed incorrectly
- ♦ Crankshaft sensor loose
- ♦ Crankshaft sensor faulty

Fault setting condition(s)

- Signal not OK (missing or implausible)

Diagnostic information - DME (DFI) control unit

Crankshaft sensor

Diagnostic conditions

- Engine start
- then driving with overrun phase(s)
- **PDK only:**
 - then 2 x automatic engine stop initiated by Start/Stop system with automatic restart

Possible fault causes

- ♦ Starter battery flat, with low ambient temperature
- ♦ Loose contact or corrosion on connectors or lines
- ♦ Phase sensor wheel faulty (damaged/bent)
- ♦ Crankshaft sensor loose
- ♦ Crankshaft sensor not installed or installed incorrectly
- ♦ Crankshaft sensor faulty

Fault setting condition(s)

- The crankshaft sensor detects too many or not enough tooth flanks on the sensor wheel, at engine start, with engine running or overrun phases

P0341

Diagnostic information - DME (DFI) control unit

Bank 1 intake camshaft sensor

Diagnostic conditions

- Engine running

Possible fault causes

- ♦ Loose contact or corrosion on connectors or lines
- ♦ Short circuit to B+/ground/between lines or open circuit in the lines
- ♦ Bank 1 intake camshaft sensor faulty

P0342

Diagnostic information - DME (DFI) control unit

Bank 1 intake camshaft sensor

Diagnostic conditions

- Engine running

Possible fault causes

- ♦ Loose contact or corrosion on connectors or lines
- ♦ Short circuit to B+/ground/between lines or open circuit in the lines
- ♦ Bank 1 intake camshaft sensor faulty

P0343

Diagnostic information - DME (DFI) control unit

Bank 1 intake camshaft sensor

Diagnostic conditions

- Engine running

Possible fault causes

- ♦ Dynamic driving style when oil level is near minimum (oil foaming)
- ♦ Loose contact or corrosion on connectors or lines
- ♦ Short circuit to B+/ground/between lines or open circuit in the lines
- ♦ Bank 1 intake camshaft sensor faulty
- ♦ Camshaft rotor (phase sensor wheel) severely damaged
- ♦ Chain tensioner loose/faulty
- ♦ Camshaft controller loose/faulty

P0344

Diagnostic information - DME (DFI) control unit

Bank 1 intake camshaft sensor

Diagnostic conditions

- Engine running

Possible fault causes

- ♦ Loose contact or corrosion on connectors or lines
- ♦ Short circuit to B+/ground/between lines or open circuit in the lines
- ♦ Bank 1 intake camshaft sensor faulty
- ♦ Camshaft adjustment faulty (locking)

P0346

Diagnostic information - DME (DFI) control unit

Bank 2 intake camshaft sensor

Diagnostic conditions

- Engine running

Possible fault causes

- ♦ Loose contact or corrosion on connectors or lines
- ♦ Short circuit to B+/ground/between lines or open circuit in the lines
- ♦ Bank 2 intake camshaft sensor faulty

P0347

Diagnostic information - DME (DFI) control unit

Bank 2 intake camshaft sensor

Diagnostic conditions

- Engine running

Possible fault causes

- ♦ Loose contact or corrosion on connectors or lines
- ♦ Short circuit to B+/ground/between lines or open circuit in the lines
- ♦ Bank 2 intake camshaft sensor faulty

P0348

Diagnostic information - DME (DFI) control unit

Bank 2 intake camshaft sensor

Diagnostic conditions

- Engine running

Possible fault causes

- ♦ Dynamic driving style when oil level is near minimum (oil foaming)
- ♦ Loose contact or corrosion on connectors or lines
- ♦ Short circuit to B+/ground/between lines or open circuit in the lines
- ♦ Bank 2 intake camshaft sensor faulty
- ♦ Camshaft rotor (phase sensor wheel) severely damaged
- ♦ Chain tensioner loose/faulty
- ♦ Camshaft controller loose/faulty

P0349

Diagnostic information - DME (DFI) control unit

Bank 2 intake camshaft sensor

Diagnostic conditions

- Engine running

Possible fault causes

- ♦ Loose contact or corrosion on connectors or lines
- ♦ Short circuit to B+/ground/between lines or open circuit in the lines
- ♦ Bank 2 intake camshaft sensor faulty
- ♦ Camshaft adjustment faulty (locking)

P034A

Diagnostic information - DME (DFI) control unit

Reference mark for crankshaft sensor not detected

Diagnostic conditions

- Engine running
- Then automatic engine stop and restart by the Start Stop system

Possible fault causes

- ♦ Loose contact or corrosion on connectors or lines
- ♦ Crankshaft sensor loose
- ♦ Crankshaft sensor faulty

Fault setting condition(s)

The phase sensor wheel (flywheel) has 58 teeth, followed by a gap that corresponds to 2 teeth.

- This fault is set if the gap is detected although either less than 56 teeth or more than 60 teeth were counted since the last gap.

Diagnostic information - DME (DFI) control unit

Bar ignition module on cylinder 1

⚠ WARNING

High voltage !

Risk of death or serious injury due to high voltage or high current!

- ♦ Start engine only if all the bar ignition modules have been installed and connected.

NOTE

Sensitive electronics!

Risk of irreparable damage to electronic components!

- ♦ **DO NOT** carry out electrical tests.
- ♦ **DO NOT** connect external power sources.
- ♦ Check ground supply.

Diagnostic conditions

- Time since engine start > 2 second(s)
- Engine temperature > -30 °C (-22 °F)

Possible fault causes

- ♦ Loose contact or corrosion on connectors or lines
- ♦ Short circuit to B+ or open circuit in the lines (ground supply)
- ♦ Short circuit to ground or open circuit in the lines (power supply)
- ♦ Short circuit to B+/ground or open circuit in the line (activation)
- ♦ Bar ignition module on cylinder 1 faulty

Diagnostic information - DME (DFI) control unit

Bar ignition module on cylinder 2

⚠ WARNING

High voltage !

Risk of death or serious injury due to high voltage or high current!

- ♦ Start engine only if all the bar ignition modules have been installed and connected.

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Diagnostic conditions

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Possible fault causes

- ♦ Loose contact or corrosion on connectors or lines
- ♦ Short circuit to B+ or open circuit in the lines (ground supply)
- ♦ Short circuit to ground or open circuit in the lines (power supply)
- ♦ Short circuit to B+/ground or open circuit in the line (activation)
- ♦ Bar ignition module on cylinder 2 faulty

Diagnostic information - DME (DFI) control unit

Bar ignition module on cylinder 3

⚠ WARNING

High voltage !

Risk of death or serious injury due to high voltage or high current!

- ♦ Start engine only if all the bar ignition modules have been installed and connected.

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Possible fault causes

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- ♦ Short circuit to B+ or open circuit in the lines (ground supply)
- ♦ Short circuit to ground or open circuit in the lines (power supply)
- ♦ Short circuit to B+/ground or open circuit in the line (activation)
- ♦ Bar ignition module on cylinder 3 faulty

Diagnostic information - DME (DFI) control unit

Bar ignition module on cylinder 4

⚠ WARNING

High voltage !

Risk of death or serious injury due to high voltage or high current!

- ♦ Start engine only if all the bar ignition modules have been installed and connected.

NOTE

Sensitive electronics!

Risk of irreparable damage to electronic components!

- ♦ **DO NOT** carry out electrical tests.
- ♦ **DO NOT** connect external power sources.
- ♦ Check ground supply.

Diagnostic conditions

- Time since engine start > 2 second(s)
- Engine temperature > -30 °C (-22 °F)

Possible fault causes

- ♦ Loose contact or corrosion on connectors or lines
- ♦ Short circuit to B+ or open circuit in the lines (ground supply)
- ♦ Short circuit to ground or open circuit in the lines (power supply)
- ♦ Short circuit to B+/ground or open circuit in the line (activation)
- ♦ Bar ignition module on cylinder 4 faulty

Diagnostic information - DME (DFI) control unit

Bar ignition module on cylinder 5

⚠ WARNING

High voltage !

Risk of death or serious injury due to high voltage or high current!

- ♦ Start engine only if all the bar ignition modules have been installed and connected.

NOTE

Sensitive electronics!

Risk of irreparable damage to electronic components!

- ♦ **DO NOT** carry out electrical tests.
- ♦ **DO NOT** connect external power sources.
- ♦ Check ground supply.

Diagnostic conditions

- Time since engine start > 2 second(s)
- Engine temperature > -30 °C (-22 °F)

Possible fault causes

- ♦ Loose contact or corrosion on connectors or lines
- ♦ Short circuit to B+ or open circuit in the lines (ground supply)
- ♦ Short circuit to ground or open circuit in the lines (power supply)
- ♦ Short circuit to B+/ground or open circuit in the line (activation)
- ♦ Bar ignition module on cylinder 5 faulty

Diagnostic information - DME (DFI) control unit

Bar ignition module on cylinder 6

⚠ WARNING

High voltage !

Risk of death or serious injury due to high voltage or high current!

- ♦ Start engine only if all the bar ignition modules have been installed and connected.

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Sensitive electronics!

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- ♦ Short circuit to B+ or open circuit in the lines (ground supply)
- ♦ Short circuit to ground or open circuit in the lines (power supply)
- ♦ Short circuit to B+/ground or open circuit in the line (activation)
- ♦ Bar ignition module on cylinder 6 faulty

P0366

Diagnostic information - DME (DFI) control unit

Bank 1 exhaust camshaft sensor

Diagnostic conditions

- Engine running

Possible fault causes

- ♦ Loose contact or corrosion on connectors or lines
- ♦ Short circuit to B+/ground/between lines or open circuit in the lines
- ♦ Bank 1 exhaust camshaft sensor faulty

P0367

Diagnostic information - DME (DFI) control unit

Bank 1 exhaust camshaft sensor

Diagnostic conditions

- Engine running

Possible fault causes

- ♦ Loose contact or corrosion on connectors or lines
- ♦ Short circuit to B+/ground/between lines or open circuit in the lines
- ♦ Bank 1 exhaust camshaft sensor faulty

P0368

Diagnostic information - DME (DFI) control unit

Bank 1 exhaust camshaft sensor

Diagnostic conditions

- Engine running

Possible fault causes

- ♦ Dynamic driving style when oil level is near minimum (oil foaming)
- ♦ Loose contact or corrosion on connectors or lines
- ♦ Short circuit to B+/ground/between lines or open circuit in the lines
- ♦ Bank 1 exhaust camshaft sensor faulty
- ♦ Camshaft rotor (phase sensor wheel) severely damaged
- ♦ Chain tensioner loose/faulty
- ♦ Camshaft controller loose/faulty

P0369

Diagnostic information - DME (DFI) control unit

Bank 1 exhaust camshaft sensor

Diagnostic conditions

- Engine running

Possible fault causes

- ♦ Loose contact or corrosion on connectors or lines
- ♦ Short circuit to B+/ground/between lines or open circuit in the lines
- ♦ Bank 1 exhaust camshaft sensor faulty
- ♦ Camshaft adjustment faulty (locking)

P0370

Diagnostic information - DME (DFI) control unit

Crankshaft sensor

Diagnostic conditions

- Engine start
- then driving with overrun phase(s)
- then automatic engine stop and restart by the Start/Stop system

Possible fault causes

- ♦ Crankshaft sensor faulty

Fault setting condition(s)

- Implausible direction of rotation signal

P0371

Diagnostic information - DME (DFI) control unit

Crankshaft sensor

Diagnostic conditions

- Engine running
- Then automatic engine stop and restart by the Start Stop system

Possible fault causes

- ♦ Loose contact or corrosion on connectors or lines
- ♦ Short circuit to B+/ground/between lines or open circuit in the lines
- ♦ Phase sensor wheel (flywheel) faulty (damaged/bent)
- ♦ Crankshaft sensor not installed or installed incorrectly
- ♦ Crankshaft sensor loose
- ♦ Crankshaft sensor faulty

Fault setting condition(s)

- Signal not OK (missing or implausible)

Diagnostic information - DME (DFI) control unit

Crankshaft sensor

Diagnostic conditions

- Engine start
- then driving with overrun phase(s)
- **PDK only:**
 - then 2 x automatic engine stop initiated by Start/Stop system with automatic restart

Possible fault causes

- ♦ Starter battery flat, with low ambient temperature
- ♦ Loose contact or corrosion on connectors or lines
- ♦ Phase sensor wheel faulty (damaged/bent)
- ♦ Crankshaft sensor loose
- ♦ Crankshaft sensor not installed or installed incorrectly
- ♦ Crankshaft sensor faulty

Fault setting condition(s)

- The crankshaft sensor detects too many or not enough tooth flanks on the sensor wheel, at engine start, with engine running or overrun phases

P0391

Diagnostic information - DME (DFI) control unit

Bank 2 exhaust camshaft sensor

Diagnostic conditions

- Engine running

Possible fault causes

- ♦ Loose contact or corrosion on connectors or lines
- ♦ Short circuit to B+/ground/between lines or open circuit in the lines
- ♦ Bank 2 exhaust camshaft sensor faulty

P0392

Diagnostic information - DME (DFI) control unit

Bank 2 exhaust camshaft sensor

Diagnostic conditions

- Engine running

Possible fault causes

- ♦ Loose contact or corrosion on connectors or lines
- ♦ Short circuit to B+/ground/between lines or open circuit in the lines
- ♦ Bank 2 exhaust camshaft sensor faulty

P0393

Diagnostic information - DME (DFI) control unit

Bank 2 exhaust camshaft sensor

Diagnostic conditions

- Engine running

Possible fault causes

- ♦ Dynamic driving style when oil level is near minimum (oil foaming)
- ♦ Loose contact or corrosion on connectors or lines
- ♦ Short circuit to B+/ground/between lines or open circuit in the lines
- ♦ Bank 2 exhaust camshaft sensor faulty
- ♦ Camshaft rotor (phase sensor wheel) severely damaged
- ♦ Chain tensioner loose/faulty
- ♦ Camshaft controller loose/faulty

P0394

Diagnostic information - DME (DFI) control unit

Bank 2 exhaust camshaft sensor

Diagnostic conditions

- Engine running

Possible fault causes

- ♦ Loose contact or corrosion on connectors or lines
- ♦ Short circuit to B+/ground/between lines or open circuit in the lines
- ♦ Bank 2 exhaust camshaft sensor faulty
- ♦ Camshaft adjustment faulty (locking)

Diagnostic information - DME (DFI) control unit

Secondary-air system, bank 1 and bank 2

Diagnostic conditions

- Perform all short tests from top to bottom, up to and including secondary air.
- or
- The diagnostic function starts under the following conditions:
 - Engine running, with:
 - Ambient temperature > 5 °C (41 °F)
 - Engine start temperature 5 ... 40 °C (41 ... 104 °F)
 - Secondary-air injection active
 - None of the following faults stored: Oxygen sensor upstream of catalytic converter, engine coolant temperature sensor, camshaft adjustment, ambient pressure, ambient temperature
 - Diagnostics is interrupted for as long as the following conditions exist:
 - Engine speed > 3,500 rpm
 - Mass air flow > 750 mg/stroke
 - Overrun phase(s)

INFORMATION

- ♦ The diagnostic function ends with the end of secondary-air injection.
- ♦ A certain number of measurements must have taken place within diagnostics, between operational readiness of the oxygen sensors and the end of secondary-air injection.
- ♦ If this is not the case, diagnostics is ended without a result.

Possible fault causes

- ♦ Air intake on secondary pump constricted/blocked
- ♦ Short circuit to B+/ground or open circuit in the secondary-air pump relay line (working circuit)
- ♦ Secondary-air pump relay faulty
- ♦ Secondary-air injection line leaking or constricted
- ♦ Secondary-air valves stuck/frozen
- ♦ Secondary-air ducts on engine constricted/coked
- ♦ Fuel-air mixture preparation not adapted
- ♦ Secondary air at oxygen sensors upstream of catalytic converter
- ♦ Catalytic converters faulty (broken)

Fault setting condition(s)

- Lambda control actual value <> setpoint deviation
 - The diagnostic function checks whether deviation of the lambda controllers is as expected upon activation.

P0418

Diagnostic information - DME (DFI) control unit

Secondary-air pump relay driver

Diagnostic conditions

- Ignition on

Possible fault causes

- ♦ Loose contact or corrosion on connectors or lines
- ♦ Short circuit to B+/ground/between lines or open circuit in lines for secondary-air pump relay activation
- ♦ Secondary-air pump relay faulty